A Classical Benchmark: S&P 500 Modeling

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Why the S&P 500?

What is the S&P 500?

- Large-cap U.S. equity benchmark
- 500 leading companies, float-adjusted market-cap weighted

Why it matters

- Common “market” proxy
- Used to judge portfolios, risk appetite, and market regimes

Why returns, not prices?

- Returns measure change
- Comparable across time: percent/log scale, not index points
- Closer to model assumptions: mean more stable than price level

Index level
S□ trends over time

→

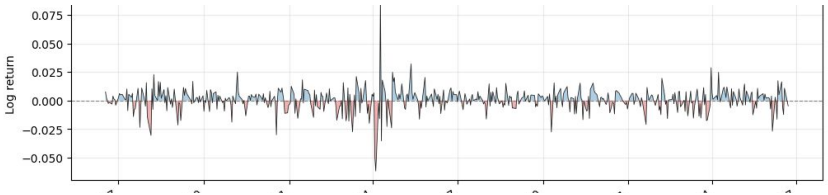
Log return
r□ = log(S□ / S□-1)

→

Model target
next day / week / month

→

Decision object



Example target: close-to-open log returns fluctuate around zero; shocks appear as spikes.

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S&P 500 Returns

$$r(t) = \log \frac{S_o(t)}{S_c(t-1)}$$

$$R(t,h) = r(t+1) + \ \ldots + r(t+h)$$

- h = 1 business day: next-session risk and direction.
- h = 5 business days: one-week tactical view.
- h = 21 business days: one-month cumulative return.

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Creating Benchmarks – Classical Models

Types	Methods	Why
Naive	Zero-return	Sanity Check
Statistical	ARIMA	Interpretable
Statistical	Kalman	Classical Methods
Machine learning	XGBoost, LightGBM	Top Performance
LLMs & agents	LLM Processes, agents	Language + reasoning

Kalman Filter

Estimate a hidden market state from noisy observations, one update at a time.

The filter repeats three moves

1 Predict → **2 Observe** → **3 Update**

Use yesterday's state estimate

Read today's market data

Blend prediction + evidence

Hidden state interpretation: bullish / steady / bearish

Instead of trusting raw noisy returns directly, the Kalman filter keeps a running belief about the underlying market state.

In market terms

State: $x_t = F_t x_{t-1} + B_t u_t + w_t$

Observation: $z_t = H_t x_t + v_t$

- x_t: hidden state
- u_t: external signals
- z_t: observed data
- w_t, v_t: shocks + noise

Key takeaway: combine yesterday's prediction with today's noisy evidence.

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Auto-regressive Models

AR(p): the lag-memory model



$$r(t) = c + \varphi_1 r_{t-1} + \varphi_2 r_{t-2} + \ldots + \varphi_p r_{t-p} + \varepsilon_t$$

Choosing p: ACF/PACF plots

ACF

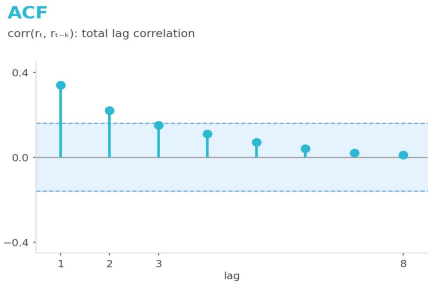
Corr(r□, r□-□) = total lag correlation

Includes direct + indirect effects.

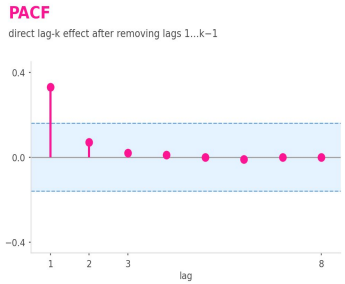
PACF

Corr(r□, r□-□ | lags 1…k−1)

Isolates the direct effect of lag k.



AR clue: often tails off / decays



Choose p: last significant spike

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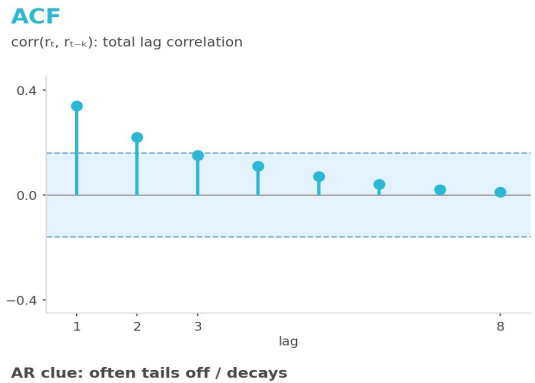
Moving Average Models

Main Idea

Today's return is influenced by recent unexpected shocks in the market.

MA(q)

$$r_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_q \varepsilon_{t-q}$$



AR-MA Models

ARMA combines lagged returns and lagged shocks; ARIMA adds differencing when needed.

$$\text{ARMA}(p, q): \quad r_t = c + \sum_i \varphi_i r_{t-i} + \varepsilon_t + \sum_t \theta_t \varepsilon_{t-t}$$

ARIMA(p, d, q): apply d differences before ARMA

ARIMA Models with Covariates

$$R_{t,t+t} = \beta_0 + \beta_r \cdot \text{lags}(r) + \beta_x \cdot X_t + \varepsilon_{t+t}$$

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Covariates

Volatility	VIX
Rates	Treasury yields, 2s10s spread
Policy	Fed funds
Macro	CPI, unemployment
Commodities	oil, gold
Risk assets	NASDAQ, dollar index

Gradient Boosted Trees: LightGBM

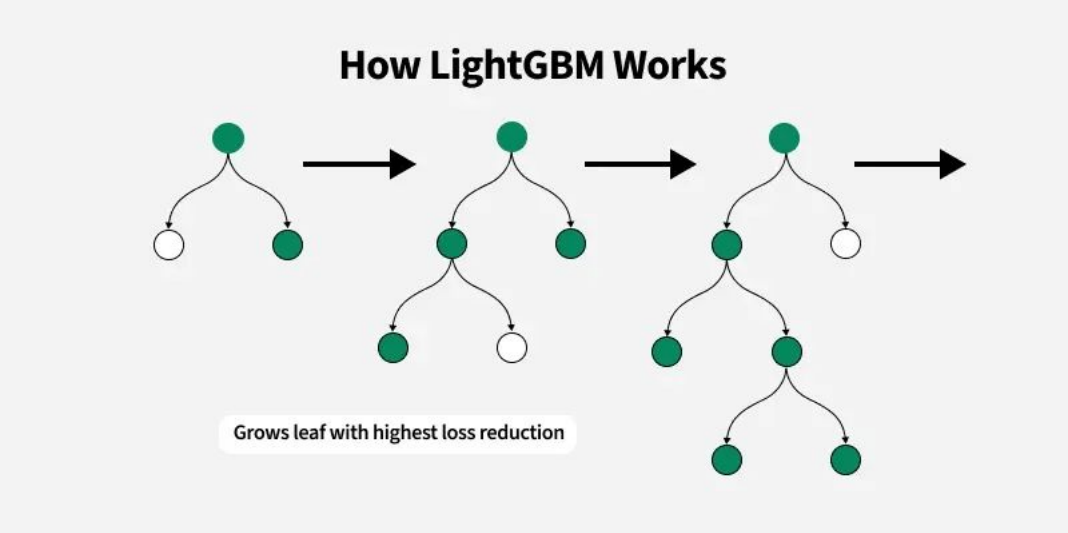
Boosted trees are the tabular benchmark for nonlinear covariate interactions.

Boosting loop



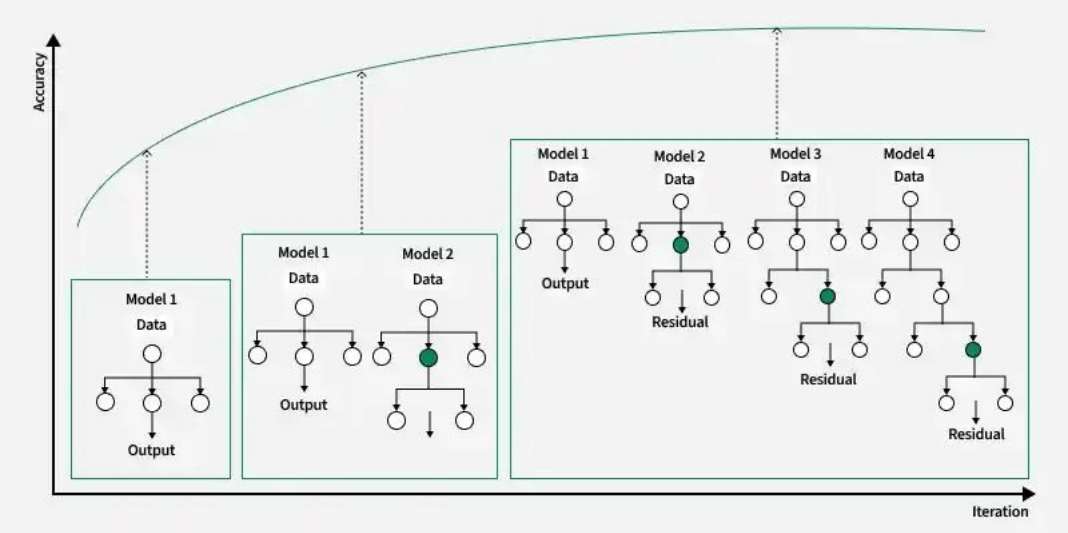
- Each new tree focuses on errors left by previous trees.
- Handles nonlinear effects and feature interactions.
- Works well with mixed-scale covariates without heavy preprocessing.

Gradient Boosted Trees: LightGBM



Source: www.geeksforgeeks.org/machine-learning/train-a-model-using-lightgbm/

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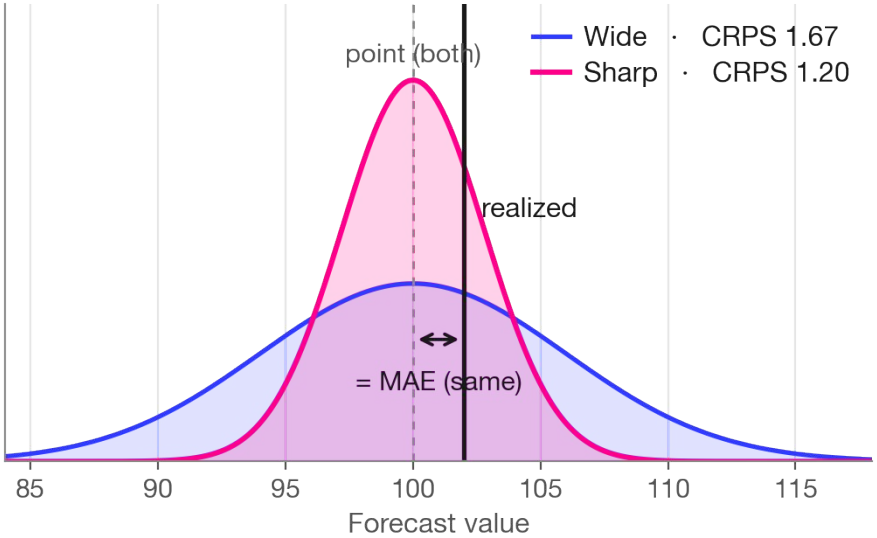
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Modeling Playbook

1	**Choose horizon**	1b, 5b, or 21b forward log return
2	**Assemble data**	price series + available covariates
3	**Create features**	lags, rolling stats, aligned covariates
4	**Fit baselines**	naive, ETS/Kalman, ARIMA
5	**Fit covariate models**	linear regression and LightGBM
6	**Backtest and select**	CRPS, direction, regime stability

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CRPS Score



Two forecasts, same point forecast — CRPS separates them, MAE can't

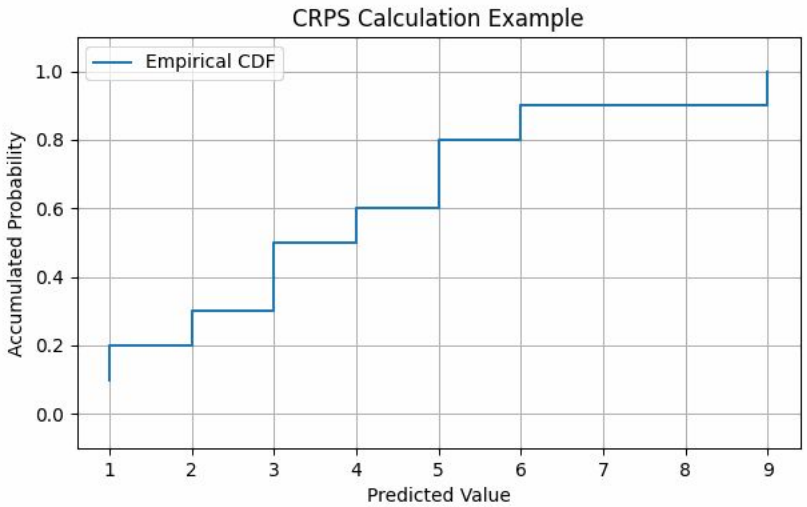
Right place, tight interval.

CRPS scores the whole distribution, not just the point — sharper, well-placed wins.

- MAE sees only the point
- CRPS rewards sharpness

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CRPS Score – Empirical



Source: skforecast.org/0.18.0/faq/probabilistic-forecasting-crps-score.html

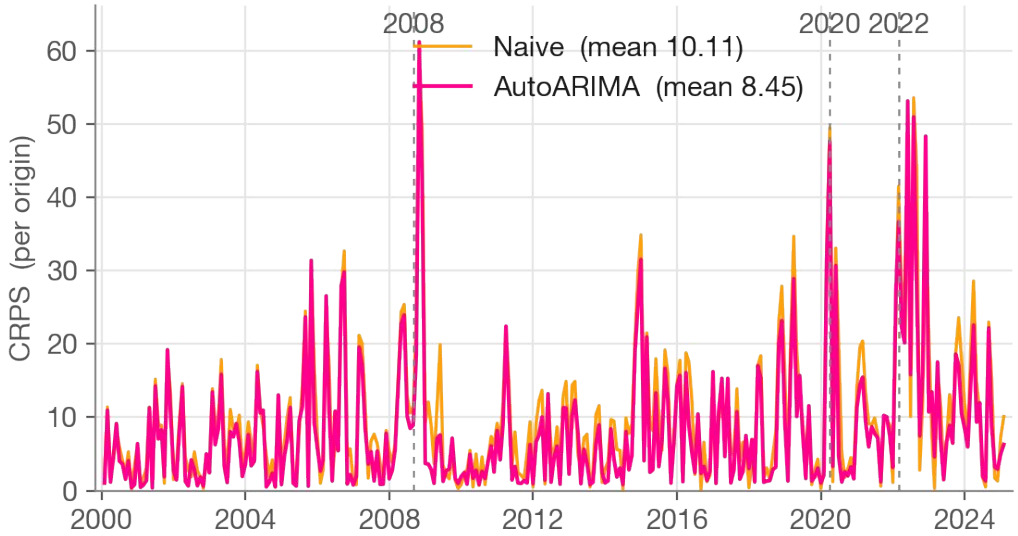
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S&P500 forecasting – Covid 19

Implemented in: implementaions/01_sp500_multivariate_backtest.ipynb

Implementation

The error clusters at the shocks



Per-origin CRPS · Naive vs AutoARIMA · 301 origins, 2000–2025

AutoARIMA wins — by ~16%.

Beats the floor — yet both spike at 2008, 2020, 2022, where context could help.

8.45

AutoARIMA · mean CRPS

10.11

Naive floor

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What to take forward

01

Honest evaluation, always

One interface, strict cutoffs, protected evals — for every method.

02

Established methods set the bar

A real classical model clearly beats the naive floor — tough to top.

03

Agents earn their place

By bringing context and reasoning at the shocks — tested, not assumed.

Up next: the conventional methods that set the bar — over to Behnoosh.